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SEMICONDUCTOR MODULE

Abstract

A semiconductor module includes a printed circuit board and a plurality of tab terminals on the printed circuit board. The plurality of tab terminals include a plurality of signal terminals and a plurality of ground terminals. The plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, and each of the plurality of tab terminals extends in a second direction transverse to the first direction. Each of the plurality of signal terminals includes a first end portion adjacent to the first edge of the printed circuit board, and an insulation member on the printed circuit board. The insulation member includes an extension portion and a plurality of protruding portions. Each of the plurality of protruding portions covers the first end portion of a respective one of the plurality of signal terminals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0033468 filed in the Korean Intellectual Property Office on Mar. 8, 2024, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

(A) Field of the Invention

[0002] The present disclosure relates to a semiconductor module.

(B) Description of the Related Art

[0003] The semiconductor module has a structure of connecting semiconductor chips and tab terminals, in that a semiconductor chip is disposed on a printed circuit board (PCB) corresponding to a substrate base, a tab terminal is disposed on a first edge of the printed circuit board, and wires are disposed on the printed circuit board. The semiconductor module may be mounted in a test socket of an automated test equipment (ATE) to perform a test for the semiconductor module, or it may be mounted in a socket of a main board to use the semiconductor module. The test socket of the automated test equipment (ATE) and the socket of the main board may include socket pins corresponding to the tab terminals of the semiconductor module, and when mounting the semiconductor module in the test socket of the automated test equipment (ATE) or mounting it in the socket of the main board, the socket pins may be electrically and physically connected to the tab terminals of the semiconductor module.

[0004] The test socket of the automated test equipment (ATE) or the socket of the main board before mounting the semiconductor module may be in a charged state, since charges moved from other electronic devices may be accumulated. While mounting the semiconductor module in the socket in the charged state, when a signal terminal among the tab terminals of the semiconductor module first contacts the socket pin of the socket, the charges accumulated in the socket move to the signal terminal of the semiconductor module through the socket pin, and move to the semiconductor chip through the signal terminal and signal wires. The socket, the socket pin, the signal terminal, and the signal wire form a discharge path, and the discharge path may damage an internal circuit of the semiconductor chip, and may be the cause for a product not operating properly, and may cause deterioration of signal integrity (SI).

SUMMARY OF THE INVENTION

[0005] In a semiconductor module including tab terminals including signal terminals and ground terminals and being mountable in a socket, when mounting a semiconductor module on a socket, an end portion of each signal terminal may be covered by an insulation member such that the ground terminals may contact the socket pin of the socket before the signal terminals.

[0006] A semiconductor module including tab terminals including signal terminals and ground terminals and being mountable in a socket, an end portion of each signal terminal and a tie-bar extending from each signal terminal may be covered by an insulation member without removing tie-bars extending from tab terminals.

[0007] A semiconductor module may include a printed circuit board, and a plurality of tab terminals on the printed circuit board, where the plurality of tab terminals may include a plurality of signal terminals and a plurality of ground terminals, where the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, where each of the plurality of tab terminals extends in a second direction that is transverse to the first direction, and

where each of the plurality of signal terminals may include a first end portion adjacent to the first edge of the printed circuit board, and a plurality of insulation members on the printed circuit board, where each of the plurality of insulation members covers the first end portion of a respective one of the plurality of signal terminals.
[0008] A semiconductor module may include a printed circuit board, a plurality of tab terminals on the printed circuit board, where the plurality of tab terminals may include a plurality of signal terminals and a plurality of ground terminals, where the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, where each of the plurality of tab terminals extends in a second direction that is transverse to the first direction, where each of the plurality of signal terminals may include first end portion adjacent to the first edge of the printed circuit board, and an insulation member on the printed circuit board, where the insulation member may include an extension portion extending in the first direction along the first edge of the printed circuit board, and a plurality of protruding portions extending in the second direction from the extension portion, and where each of the plurality of protruding portions covers the first end portion of a respective one of the plurality of signal terminals.
[0009] A semiconductor module may include a printed circuit board, a plurality of semiconductor chips on the printed circuit board, a plurality of tab terminals on the printed circuit board, where the plurality of tab terminals may include a plurality of signal terminals electrically connected to the plurality of semiconductor chips, and a plurality of ground terminals connected to a ground, where the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, where each of the plurality of tab terminals extends in a second direction that is transverse to the first direction and where each of the plurality of signal terminals may include a first end portion adjacent to the first edge of the printed circuit board, a plurality of first tie-bars on the printed circuit board, where each of the plurality of first tie-bars extends from a respective one of the plurality of signal terminals, a plurality of second tie-bars on the printed circuit board, where each of the plurality of second tie-bars extends from a respective one of the plurality of ground terminals, and a plurality of insulation members on the printed circuit board, where each of the plurality of insulation members covers the first end portion and the first tie-bar of a respective one of the plurality of signal terminals.
[0010] An end portion of each signal terminal may be covered with an insulation member such that the ground terminals may contact the socket pin of the socket before the signal terminals when mounting a semiconductor module in a socket. Accordingly, when mounting a semiconductor module in a socket, a discharge path along which the charge accumulated in the socket may move into a semiconductor chip through the socket, the socket pin, the signal terminal, and the signal wire, and potentially damage a circuit in the semiconductor chip may be removed, and the charge accumulated in the socket may be discharged to the ground through the ground terminals.
[0011] Without removing tie-bars extending from tab terminals, the end portion of each signal terminal and the tie-bar extending from each signal terminal may be covered with an insulation member. Accordingly, in the manufacturing process of the semiconductor module, a complex etching back process for removing tie-bars extending from tab terminals may be omitted, such that the turnaround time (TAT) of the semiconductor module manufacturing process may be reduced, and the manufacturing cost of the semiconductor module may be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is a perspective view showing a semiconductor module aligned on a socket, in order to be mounted in the socket.
[0013] FIG. **2** is a cross-sectional view sequentially showing a process in which the semiconductor

module of FIG. **1** is mounted in the socket of FIG. **1**. The semiconductor module and socket of FIG. **2** is shown based on a cross-section of the semiconductor module and socket of FIG. **1** along line B-B′.

[0014] FIG. **3** is a perspective view showing a semiconductor module mounted in the socket.

[0015] FIG. **4** is an enlarged perspective view showing that tab terminals of region A of the semiconductor module of FIG. **1** are in contact with a socket pin of a socket when a semiconductor module is mounted in the socket.

[0016] FIG. **5** is a perspective view showing a semiconductor module of an embodiment.

[0017] FIG. **6** is an enlarged perspective view showing a region C of the semiconductor module of FIG. **5**.

[0018] FIG. **7** is a cross-sectional view of the semiconductor module of FIG. **6** taken along line D-D′ and line E-E′.

[0019] FIG. **8** is a perspective view showing that tab terminals of region C of the semiconductor module of FIG. **5** are in contact with a socket pin of a socket when a semiconductor module of an embodiment is mounted in the socket.

[0020] FIG. **9** is a perspective view showing an exemplary variation of the semiconductor module of FIG. **5**.

[0021] FIG. **10** is an enlarged perspective view showing a region F of the semiconductor module of FIG. **9**.

[0022] FIG. **11** is a perspective view showing a semiconductor module of another embodiment.

[0023] FIG. **12** is an enlarged perspective view showing a region G of the semiconductor module of FIG. **11**.

[0024] FIG. **13** is a cross-sectional view of the semiconductor module of FIG. **12** taken along line H-H′ and line I-I′.

[0025] FIG. **14** is a perspective view showing that tab terminals of region G of the semiconductor module of FIG. **11** are in contact with a socket pin of a socket when a semiconductor module of an embodiment is mounted in the socket.

[0026] FIG. **15** is a perspective view showing an exemplary variation of the

[0027] semiconductor module of FIG. **11**.

[0028] FIG. **16** is an enlarged perspective view showing a region J of the semiconductor module of FIG. **15**.

[0029] FIG. **17** is a perspective view showing a semiconductor module of still another embodiment.

[0030] FIG. **18** is an enlarged perspective view showing a region K of the semiconductor module of FIG. **17**.

[0031] FIG. **19** is a cross-sectional view of the semiconductor module of FIG. **18** taken along line L-L′ and line M-M′.

[0032] FIG. **20** is a perspective view showing that tab terminals of region K of the semiconductor module of FIG. **17** are in contact with a socket pin of a socket when a semiconductor module of an embodiment is mounted in the socket.

[0033] FIG. **21** is a perspective view showing an exemplary variation of the semiconductor module of FIG. **17**.

[0034] FIG. **22** is an enlarged perspective view showing a region N of the semiconductor module of FIG. **21**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0035] Hereinafter, embodiments of the disclosure will be hereinafter described in detail with reference to the accompanying drawings, such that those skilled in the art may easily implement time. The disclosure may be implemented in various forms, and may not necessarily limited to embodiments described herein.

[0036] The drawings and description are to be regarded as illustrative in nature and not restrictive.

Like reference numerals designate like elements throughout the specification.

[0037] Further, in the drawings, the size and thickness of each element may be arbitrarily illustrated for ease of description, and the present disclosure is not necessarily limited to those illustrated in the drawings.

[0038] Throughout this specification and the claims that follow, when it is described that an element is “coupled or connected” to another element, the element may be “directly coupled or connected” to the other element or “indirectly coupled or connected” to the other element through a third element. In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0039] It will be understood that when an element such as a layer, film, region, area, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. Further, in the specification, the word “on” or “above” means positioned on or below the object portion, and does not necessarily mean positioned on the upper side of the object portion based on a gravitational direction.

[0040] Further, throughout the specification, the phrase “in a plan view” or “on a plane” means viewing a target portion from the top, and the phrase “in a cross-sectional view” or “on a cross-section” means viewing a cross-section formed by vertically cutting a target portion from the side.

[0041] Hereinafter, a semiconductor module **10** of an embodiment will be described with reference to the drawings.

[0042] FIG. **1** is a perspective view showing the semiconductor module **10** aligned with the socket **20**, in order to be mounted in a socket **20**.

[0043] Referring to FIG. **1**, the semiconductor module **10** may include a printed circuit board (PCB) **110**, semiconductor chips **120**, tab terminals **130**, and tie-bars **132**E (tie-bars of which a part is removed by an etch back process; refer to FIG. **4**). In an embodiment, the semiconductor module **10** may include a memory module. In an embodiment, the semiconductor module **10** may include a dual In-line memory module (DIMM), small outline dual in-line memory module (SODIMM), unbuffered dual inline memory module (UDIMM), registered dual inline-memory module (RDIMM), load-reduced dual inline-memory module (LRDIMM), hypercloud dual inline-memory module (HDIMM), non-volatile DIMM (NVDIMM), fully buffered dual inline-memory module (FB-DIMM), CXL memory module (CMM), multi-ranked buffered DIMM (MRDIMM), or low-power compression attached memory module (LPCAMM).

[0044] The printed circuit board (PCB) **110** may be a substrate on which the semiconductor chips **120** are mounted. In an embodiment, the printed circuit board (PCB) **110** may have a multi-layer structure in which core layers and prepregs are alternately stacked, and a solder mask stacked in an uppermost portion and a lowermost portion. In an embodiment, core layers may include FR-4 and a metal trace attached to top and bottom sides of the FR-4.

[0045] Within the multi-layer structure, the printed circuit board (PCB) **110** may include a ground wire layer, a power wire layer, and a signal wire layer. The ground wire layer may be a bulk layer connected to the ground. The power wire layer may be a bulk layer connected to a power supply, and providing a path for supplying electric power the power supply. The signal wire layer may be a layer in which the wire patterns are formed. The ground wire layer, the power wire layer, and the signal wire layer may be disposed at different levels, and may be connected through vias.

[0046] The semiconductor chips **120** may be mounted on a surface of the printed circuit board (PCB) **110**. The semiconductor chips **120** may be mounted on one surface or both surfaces of the printed circuit board (PCB) **110**. In an embodiment, the semiconductor chips **120** may be linearly arranged along a first direction (X direction). In an embodiment, the semiconductor chips **120** may be mounted on the printed circuit board (PCB) **110** by a flip chip method. In an embodiment, a semiconductor chip **120** may be a memory chip or logic chip. In an embodiment, when a semiconductor chip **120** is a memory chip, the semiconductor chip **120** may include dynamic random-access memory (DRAM), static random-access memory (SRAM), Flash Memory, erasable programmable read-only memory (EPROM), phase change random-access memory (PRAM), magnetic random-access memory (MRAM), resistive random-access memory (RRAM), or spin-transfer torque magnetic random-access memory (STT-MRAM).

[0047] The tab terminals **130** may be disposed along a first edge **110**E of the printed circuit board (PCB) **110**. The tab terminals **130** may be disposed on both surfaces of the printed circuit board (PCB) **110**. The tab terminals **130** may be arranged in the first direction (X direction) along a first edge of the printed circuit board (PCB) **110**. Each of the tab terminals **130** may be spaced apart from the neighboring tab terminal **130** at a predetermined interval in the first direction (X direction). Each of the tab terminals **130** may extend in a second direction (Y direction) that is transverse to the first direction (X direction). In an embodiment, the tab terminal **130** may be formed by performing an electroplating process or electroless plating process. In an embodiment, the tab terminal **130** may include copper (Cu), nickel (Ni), or gold (Au).

[0048] The tab terminals **130** may include signal terminals **141**, ground terminals **151**, and power terminals (not shown). The signal terminals **141** may include a data terminal where data is input or output, an address terminal where an address signal is input, a command terminal where a command signal is input, a clock terminal where a clock signal is input, and a control terminal where a control signal is input. The signal terminals **141** may be electrically connected to the signal wire layer within the multi-layer structure of the printed circuit board (PCB) **110** and a signal wire line on a surface of the printed circuit board (PCB) **110**. The ground terminals **151** may be electrically connected to the ground wire layer within the multi-layer structure of the printed circuit board (PCB) **110**. Power terminals may be electrically connected to the power wire layer within the multi-layer structure of the printed circuit board (PCB) **110**. In an embodiment, the tab terminals **130** may further include plating terminals or no-connection terminals.

[0049] Tie-bars **132**E (refer to FIG. **4**) may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**. The tie-bars **132**E may be disposed on both surfaces of the printed circuit board (PCB) **110**. Each of the tie-bars **132**E may extend from a first end of each of the tab terminals **130** toward the first edge **110**E of the printed circuit board (PCB) **110**. A first end of the tab terminal **130** may be defined as, among both ends of the tab terminal **130**, an end at a location adjacent to the first edge **110**E of the printed circuit board (PCB) **110**. A second end of the tab terminal **130** may be defined as, among the both ends of the tab terminal **130**, an end at a location distal to the first edge **110**E of the printed circuit board (PCB) **110**. The tie-bars **132**E may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**. Each of the tie-bars **132**E may be spaced apart from the neighboring tie-bar **132**E at a predetermined interval in the first direction (X direction). Each of the tie-bars **132**E may extend in the second direction (Y direction) that is transverse to the first direction (X direction).

[0050] The tie-bars **132**E may serve to perform electroplating on the tab terminals **130** and wires. After the electroplating is performed, the tie-bars **132**E are cut together while the flat plate pf the printed circuit board (PCB) **110** is cut, and in this case, when a line width of the tie-bars **132**E in the first direction (X direction) is large, the tab terminals **130** may be damaged during the cutting process. Therefore, the tie-bar **132**E may have a line width in the first direction (X direction) smaller than a line width of the tab terminal **130** in the first direction (X direction). Each of the tie-bars **132**E may extend from a leftmost side portion, a rightmost side portion, or a portion between the leftmost side and the rightmost side of the first end of each of the tab terminals **130**. In an embodiment, the tie-bar **132**E may include copper (Cu), nickel (Ni), or gold (Au).

[0051] The semiconductor module **10** may include wires connecting the semiconductor chips **120**, other elements (not shown), and the tab terminals **130**. These wires may have any pattern on the printed circuit board (PCB) **110**.

[0052] The socket **20** may be located on an automated test equipment (ATE) or a main board. The semiconductor module **10** may be mounted in the socket **20**, and the socket **20** may couple and support the semiconductor module **10**. The socket **20** may electrically connect the automated test equipment (ATE) and the semiconductor module **10** or the main board and the semiconductor module **10**.

[0053] The socket **20** may include a frame **210**, an insertion portion **220**, and socket pins **230** (see FIG. **2**). The frame **210** may support the semiconductor module **10**. The frame **210** may be formed of an insulating material. In an embodiment, it may include plasma enhanced oxide (PEOX), tetraethyl orthosilicate (TEOS), boro tetraethyl orthosilicate (BTEOS), phosphorous tetraethyl orthosilicate (PTEOS), boro phospho tetraethyl orthosilicate (BPTEOS), boro silicate glass (BSG), phospho silicate glass (PSG), or boro-phospho-silicate glass (BPSG).

[0054] The insertion portion **220** may have a recessed structure such that the semiconductor module **10** inserted therein. A lower portion of the semiconductor module **10** may be inserted into the insertion portion **220**. The tab terminals **130** of the semiconductor module **10** may be inserted into the insertion portion **220**.

[0055] The socket pins **230** may be located on an inner side surface of the insertion portion **220** (see FIG. **2**). The socket pins **230** may be arranged along the first direction (X direction). Each of the socket pins **230** may be spaced apart from the neighboring socket pin **230** at a predetermined interval in the first direction (X direction). Each of the socket pins **230** may extend in the second direction (Y direction) that is transverse to the first direction (X direction). Each of the socket pins **230** may correspond to each of the tab terminals **130**. In an embodiment, the socket pin **230** may include at least one of copper (Cu), aluminum (AI), nickel (Ni), silver (Ag), gold (Au), platinum (Pt), tin (Sn), lead (Pb), titanium (Ti), chromium (Cr), palladium (Pd), indium (In), zinc (Zn) and carbon (C), or may include an alloy containing at least one thereof.

[0056] FIG. **2** is a cross-sectional view sequentially showing a process in which the semiconductor module **10** is mounted in the socket **20**. The semiconductor module **10** and the socket **20** of FIG. **2** is shown based on a cross-section of the semiconductor module **10** and the socket **20** of FIG. **1** along line B-B′.

[0057] Referring to FIG. **2**, a step (a) in which the semiconductor module **10** is aligned in the socket **20**, a step (b) in which the semiconductor module **10** enters the insertion portion **220** of the socket **20** and starts to contact the socket pin **230**, and a step (c) in which the semiconductor module **10** is mounted within the insertion portion **220** of the socket **20** may be sequentially performed.

[0058] The step (a) may be a step for aligning the semiconductor module **10** on the socket **20**, in order to mount the semiconductor module **10** in the socket **20**. The socket **20** may include a fixing portion **221** within the frame **210**. The fixing portion **221** may secure the socket pin **230** to the frame **210**. The socket pin **230** may extend in the second direction (Y direction). The socket pin **230** may include a first extension portion **231**, a contact portion **232**, and a second extension portion **233**. A lower portion of the first extension portion **231** may be secured to the frame **210** by the fixing portion **221**, and may be electrically connected to the automated test equipment (ATE) or the main board. As it goes upward, an upper portion of the first extension portion **231** may extend toward an insertion portion **220** of the socket **20**. The contact portion **232** may be a portion to be in contact with the tab terminal **130** of the semiconductor module **10** mounted in the insertion portion **220** of the socket **20**. As it goes upward, the second extension portion **233** may extend toward an exterior of the frame **210**. The socket pins **230** may have elasticity (i.e., the socket pins **230** may flex or move during insertion of a semiconductor module **10** and then return to an unflexed state when the semiconductor module **10** is removed).

[0059] The step (b) may be a step in which the semiconductor module **10** enters the insertion portion **220** of the socket **20** and starts to contact the socket pin **230**. The semiconductor module **10** may enter the insertion portion **220** while pushing the socket pin **230**. Since the socket pin **230** has elasticity, the socket pin **230** may be pushed to move in an exterior direction of the frame **210** (represented by opposing arrows in FIG. **2**), by the entrance of the semiconductor module **10**. When the semiconductor module **10** starts to contact the contact portion **232**, the contact portion **232** may start the contact at a lowermost portion of each of the tab terminals **130** of the semiconductor module **10**.

[0060] The step (c) may be a step in which the entrance of the semiconductor module **10** into the insertion portion **220** of the socket **20** is completed. Since the socket pins **230** has elasticity, by the socket pins **230** being pushed to move in the exterior direction of the frame **210**, the elastic force may act in an interior direction of the frame **210**, such that the socket pins **230** may fix (i.e. grip and secure) the semiconductor module **10** within the frame **210**.

[0061] In the present disclosure, although the socket **20** including the fixing portion **221** within the frame **210** and the socket pin **230** including the first extension portion **231**, the contact portion **232**, and the second extension portion **233** are illustrated and described, it is not limited thereto, and the socket **20** and the socket pin **230** having various shapes and structures may be included in the scope of the present disclosure.

[0062] FIG. **3** is a perspective view showing the semiconductor module **10** mounted in the socket **20**.

[0063] Referring to FIG. **3**, the semiconductor module **10** may be mounted in the socket **20**. The frame **210** of the socket **20** may support the semiconductor module **10**. The semiconductor module **10** may be located within the insertion portion **220** of the socket **20**. The socket pins **230** may fix the semiconductor module **10**. The socket pins **230** may electrically connect the semiconductor module **10** to the automated test equipment (ATE) or the main board. Each of the socket pins **230** may correspond to each among the tab terminals **130** of the semiconductor module **10**.

[0064] FIG. **4** is a perspective view showing that the tab terminals **130** of a region A of the semiconductor module **10** of FIG. **1** are in contact with the socket pin **230** of the socket **20** when the semiconductor module **10** is mounted in the socket **20**.

[0065] The test socket of the automated test equipment (ATE) **20** or the socket **20** of the main board before mounting being the semiconductor module **10** may be in a charged state that the charge e moved from other electronic devices are accumulated. As such, when mounting the semiconductor module **10** in the socket **20** in the charged state, if the signal terminal **141** among the tab terminals **130** of the semiconductor module **10** first contacts the socket pin **230** of the socket **20** in the charged state, the charge e accumulated in the socket **20** may move to the signal terminal **141** of the semiconductor module **10** through the socket pin **230** in contact with the signal terminal **141**, and may move to the semiconductor chip **120** via a signal wire **143** through the signal terminal **141**. The socket **20**, the socket pin **230**, the signal terminal **141**, and the signal wire **143** may form a discharge path, and the discharge path may damage a circuit within the semiconductor chip **120**, which may be a cause to prevent the product from operating, or to deteriorate signal integrity (SI).

[0066] FIG. **5** is a perspective view showing the semiconductor module **10** of an embodiment.

[0067] Referring to FIG. **5**, the semiconductor module **10** may include insulation members **160**. The insulation members **160** may be disposed on the printed circuit board (PCB) **110**. Each of the insulation members **160** may cover a first end portion of each of the signal terminals **141** and a first tie-bar **142** (see FIG. **6**) that extends from the first end portion and is not removed by the etch back process. The first end portion may be defined as a surface becoming to contact the socket pin **230** when the insulation member **160** is not present, located adjacent to the first edge **110**E of the printed circuit board (PCB) **110**, and having a predetermined area. The insulation members **160** may not cover a ground terminal, a power terminal, plating terminals, or no-connection terminals.

[0068] The insulation members **160** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**. The insulation members **160** may be disposed on both surfaces of the printed circuit board (PCB) **110**. The insulation members **160** may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**. Each of the insulation members **160** may be spaced apart from the neighboring insulation member **160** at a predetermined interval in the first direction (X direction).

[0069] FIG. **6** is an enlarged perspective view showing a region C of the semiconductor module **10** of FIG. **5**.

[0070] Referring to FIG. **6**, the semiconductor module **10** may include the signal terminals **141** and the ground terminals **151** corresponding to a part of the tab terminals **130**. The signal terminals **141** and the ground terminals **151** may be disposed on the printed circuit board (PCB) **110**. The signal terminals **141** and the ground terminals **151** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**. The signal terminals **141** and the ground terminals **151** may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**. The signal terminal **141** may be spaced apart from the ground terminal **151** at a predetermined interval in the first direction (X direction). The signal terminals **141** and the ground terminals **151** may extend in the second direction (Y direction) that is transverse to the first direction (X direction). Each of the signal terminal **141** may have a first end and a second end. The first end may be defined as, among both ends of the signal terminal **141**, an end at a location adjacent to the first edge **110**E of the printed circuit board (PCB) **110**. The second end may be defined as, among the both ends of the signal terminal **141**, an end at a location distal to the first edge **110**E of the printed circuit board (PCB) **110**.

[0071] The semiconductor module **10** may include the tie-bars **132** that are not removed by the etch back process. The tie-bars **132** may include first tie-bars **142** and second tie-bars **152**. The first tie-bars **142** and the second tie-bars **152** may be disposed on the printed circuit board (PCB) **110**. The first tie-bar **142** may extend from a first end of the signal terminal **141** toward the first edge **110**E of the printed circuit board (PCB) **110**, to the first edge **110**E of the printed circuit board (PCB) **110**. The second tie-bar **152** may extend from a first end of the ground terminal **151** toward the first edge **110**E of the printed circuit board (PCB) **110**, to the first edge **110**E of the printed circuit board (PCB) **110**. The first tie-bars **142** and the second tie-bars **152** may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**. The first tie-bar **142** may be spaced apart from the second tie-bar **152** at a predetermined interval in the first direction (X direction). The first tie-bars **142** and the second tie-bars **152** may extend in the second direction (Y direction) that is transverse to the first direction (X direction). The first tie-bar **142** may have a line width in the first direction (X direction) smaller than a line width of the signal terminal **141** in the first direction (X direction). The second tie-bar **152** may have a line width in the first direction (X direction) smaller than a line width of the ground terminal **151** in the first direction (X direction). The first tie-bar **142** may extend from a leftmost side portion, a rightmost side portion, or a portion between the leftmost side and rightmost side of the first end of the signal terminal **141**. The second tie-bar **152** may extend from a leftmost side portion, a rightmost side portion, or a portion between the leftmost side and rightmost side of the first end of the ground terminal **151**.

[0072] In the manufacturing process of the semiconductor module **10**, the etch back process may be performed in order to remove the tie-bars **132** extending from the tab terminals **130**. The etch back process may be formed by complex processes of forming a photoresist, forming a photoresist pattern by exposing and developing the formed photoresist, etching tie-bars, and removing the photoresist pattern. The etch back process is performed through various processes, such that the turnaround time (TAT) is long, and the manufacturing cost of the semiconductor module is increased.

[0073] According to the present disclosure, without removing the tie-bars **132** extending from the tab terminals **130**, the first tie-bar **142** extending from a first end portion of each signal terminal **141** may be covered with the insulation member **160**. Accordingly, in the manufacturing process of the semiconductor module **10**, the complex etch back process for removing the tie-bars **132** extending from the tab terminals **130** may be omitted. Therefore, the turnaround time (TAT) required for manufacturing the semiconductor module **10** may be reduced, and may reduce manufacturing cost of the semiconductor module **10**.

[0074] The semiconductor module **10** may include signal wires **143**. The signal wires **143** may be disposed on the printed circuit board (PCB) **110**. The signal wire **143** may extend from a second end of the signal terminal **141** in a direction toward the semiconductor chip **120**.
[0075] The semiconductor module **10** may include ground wires **153**. The ground wires **153** may be disposed on the printed circuit board (PCB) **110**. The ground wire **153** may extend from a second end of the ground terminal **151** in a direction toward a via hole **154**. Via holes **154** may be connected to the ground wire layer within the printed circuit board (PCB) **110**.
[0076] The semiconductor module **10** may include the insulation members **160**. The insulation members **160** may be disposed on the printed circuit board (PCB) **110**. Each of the insulation members **160** may cover the first end portion of each of the signal terminals **141** and the first tie-bar **142** that is not removed by the etch back process. The insulation members **160** may not cover a ground terminal, a power terminal, plating terminals or no-connection terminals. The insulation members **160** may be spaced apart from the ground terminal, power terminal, plating terminals or no-connection terminals. The insulation members **160** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**, as illustrated. The insulation members **160** may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**.
[0077] Each of the insulation members **160** may include a first region R**1** and a second region R**2** defined by dividing a plane of each insulation member **160**. The first region R**1** of the insulation member **160** may contact the printed circuit board (PCB) **110**. The second region R**2** of the insulation member **160** may contact the signal terminal **141**. The second region R**2** of the insulation member **160** may be on and cover the first tie-bar **142** that is not removed by the etch back process and the first end portion of the signal terminal **141**.
[0078] Each of the signal terminals **141** may include a first surface exposed to the air (i.e., the first surface is exposed to the external environment and not covered by anything). Each of the ground terminals **151** may include a second surface exposed to the air (i.e., the second surface is exposed to the external environment and not covered by anything). The first surface and the second surface may be a flat surface defined by the first direction (X direction) and the second direction (Y direction). Since the first end portion of the signal terminal **141** is covered by the insulation member **160** and the ground terminal **151** is not covered by the insulation member **160**, the area of the first surface may be smaller than the area of the second surface. A first length D**1** in the second direction (Y direction) from the first edge **110**E of the printed circuit board (PCB) **110** to the exposed first surface is longer than a second length D**2** in the second direction (Y direction) from the first edge **110**E of the printed circuit board (PCB) **110** to the exposed second surface.
[0079] The insulation members **160** may be formed of a material that may withstand the force when inserting the semiconductor module **10** into the socket **20**, and may not cause a defect (e.g., appearance change of the insulation member **160**, crack occurrence of the insulation member **160**, or exposing the signal terminal due to losing of at least a portion of the insulation member **160**) when repeatedly inserting the semiconductor module **10** into the socket **20**. In an embodiment, the insulation members **160** may include a photo solder resist (PSR). The PSR may be an insulation material used in the manufacturing process of the printed circuit board (PCB) **110**, and configured to coat the surface of the printed circuit board (PCB) **110** to protect the circuit. In an embodiment, the insulation members **160** may include an overcoating material. The overcoating material may be a material that may improve mechanical strength by being applied to the device to prevent damage. In an embodiment, overcoating material may include polyurethane acrylate, urethane acrylate, acrylic, modified acrylate, or epoxy.
[0080] FIG. **7** is a cross-sectional view of the semiconductor module **10** of FIG. **6** taken along line D-D′ and line E-E′. Section (A) shows a cross-section of the semiconductor module **10** of FIG. **6** taken along line D-D′, and section (B) shows a cross-section of the semiconductor module **10** of FIG. **6** taken along line E-E′.
[0081] Referring to FIG. **7**, in section (A), the signal terminal **141**, the signal wire **143** and the

insulation member **160** may be disposed on the printed circuit board (PCB) **110**. The insulation member **160** may include the first region R**1** covering the printed circuit board (PCB) **110** and the second region R**2** covering a first end portion of the signal terminal **141**. The first region R**1** of the insulation member **160** may be located at a first level, and the second region R**2** of the insulation member **160** may be located at a second level that is higher than the first level. With reference to a contact surface of the printed circuit board (PCB) **110** and the insulation member **160**, the surface of the insulation member **160** may have a first height H**1** in the first region R**1**, and the surface of the insulation member **160** may have a second height H**2** higher than the first height H**1** in the second region R**2**. In other words, a surface of the first region is located at a first level relative to the contact surface of the PCB **110**, a surface of the second region is located at a second level relative to the contact surface of the PCB **110**, and the second level is further outward (i.e., along the Z direction) from the contact surface of the PCB **110** than the first level.

[0082] In section (B), the signal terminal **141**, the first tie-bar **142** that is not removed by the etch back process, and the insulation member **160** may be disposed on the printed circuit board (PCB) **110**. The insulation member **160** may include the second region R**2** that covers the first end portion of the signal terminal **141** and the first tie-bar **142** that are not removed by the etch back process.

[0083] FIG. **8** is a perspective view showing that the tab terminals **130** of the region C of the semiconductor module **10** of FIG. **5** are in contact with the socket pin **230** of the socket **20** when the semiconductor module **10** of an embodiment is mounted in the socket **20**.

[0084] Referring to FIG. **8**, the first end portion of the signal terminal **141** may be covered by the insulation member **160**, and the ground terminal **151** may not be covered by the insulation member **160**. Therefore, when mounting the semiconductor module **10** in the socket **20** in the charged state, the ground terminal **151** or the second tie-bar **152** that is not cover by the insulation member **160** may contact the socket pin **230** of the socket **20** earlier than the signal terminal **141** of which the first end portion is cover by the insulation member **160**. The charge e accumulated in the socket **20** may move to the ground terminal **151** of the semiconductor module **10** through the socket pin **230** in contact with the ground terminal **151** or the second tie-bar **152**, move to the via hole **154** via the ground wire **153** through the ground terminal **151**, and exit to the ground through the ground wire layer of the printed circuit board (PCB) **110**. The discharge path may be formed as a path passing through the socket **20**, the socket pin **230**, the ground terminal **151** (or the ground terminal **151** through the second tie-bar **152**), the ground wire **153**, the via hole **154**, the ground wire layer, and the ground. Therefore, according to the present disclosure, since the discharge path leading to the ground is formed, the charge e accumulated in the socket **20** may be prevented from moving to the semiconductor chip **120**, thereby damaging a circuit within the semiconductor chip **120**, and deteriorating the signal integrity (SI).

[0085] FIG. **9** is a perspective view showing an exemplary variation of the semiconductor module **10** of FIG. **5**. FIG. **10** is an enlarged perspective view showing a region F of the semiconductor module **10** of FIG. **9**.

[0086] Referring to FIG. **9** and FIG. **10**, the semiconductor module **10** may include an insulation member **161**. The insulation member **161** may be disposed on the printed circuit board (PCB) **110**. The insulation member **161** may not cover a ground terminal, a power terminal, plating terminals or no-connection terminals. The insulation member **161** may be spaced apart from the ground terminal, power terminal, plating terminals or no-connection terminals. The insulation member **161** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**.

[0087] The insulation member **161** may include an extension portion **161**E and protruding portions **161**P. The extension portion **161**E may extend in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. The extension portion **161**E may be located between the tab terminals **130** and the first edge **110**E of the printed circuit board (PCB) **110**. The extension portion **161**E may contact the printed circuit board (PCB) **110** and the tie-bars **132** that are not removed by the etch back process. The extension portion **161**E may cover a portion of each of the tie-bars **132** that are not removed by the etch back process. The extension portion **161**E may be spaced apart from the tab terminals **130**. The extension portion **161**E may be spaced apart from the signal terminals **141** and the ground terminals **151**.

[0088] The protruding portions **161**P may extend in the second direction (Y direction) from the extension portion **161**E. The protruding portions **161**P may be integrally formed with the extension portion **161**E. The protruding portions **161**P may be disposed with a step in the second direction (Y direction) with respect to the extension portion **161**E (i.e., the protruding portions **161**P extend above the extension portions **161**E along the Y direction). The protruding portions **161**P may be arranged in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. Each of the protruding portions **161**P may contact the printed circuit board (PCB) **110**, the signal terminal **141**, and the first tie-bar **142** that is not removed by the etch back process. Each of the protruding portions **161**P may cover a first end portion of the signal terminal **141** and a portion of the first tie-bar **142** that is not removed by the etch back process. The first end portion may be defined as a surface becoming to contact the socket pin **230** when the insulation member **160** is not present, located adjacent to the first edge **110**E of the printed circuit board (PCB) **110**, and having a predetermined area. The protruding portions **161**P may be spaced apart from the ground terminals **151**. Each of the protruding portions **161**P may be spaced apart from the neighboring protruding portion **161**P at a predetermined interval in the first direction (X direction). Each of the protruding portions **161**P may include the first region R**1** and the second region R**2** defined by dividing a plane of the protruding portion **161**P. The first region R**1** may contact the printed circuit board (PCB) **110**. The second region R**2** may contact the first end portion of the signal terminal **141** and the first tie-bar **142** that is not removed by the etch back process.

[0089] FIG. **11** is a perspective view showing the semiconductor module **10** of another embodiment. FIG. **12** is an enlarged perspective view showing a region G of the semiconductor module **10** of FIG. **11**.

[0090] Referring to FIG. **11** and FIG. **12**, the semiconductor module **10** may include the insulation members **160**. Each of the insulation members **160** may cover a first end portion of each of the signal terminals **141** and a first tie-bar **142**E that extends from the first end portion and of which a portion is removed by the etch back process.

[0091] The semiconductor module **10** may include a plurality of tie-bars **132**E of which a portion is removed by the etch back process. The tie-bars **132**E of which a portion is removed by the etch back process may include the first tie-bars **142**E and the second tie-bars **152**E. The first tie-bars **142**E and the second tie-bars **152**E may be disposed on the printed circuit board (PCB) **110**. The first tie-bar **142**E may extend from the first end of the signal terminal **141** toward the first edge **110**E of the printed circuit board (PCB) **110**. The second tie-bar **152**E may extend from the first end of the ground terminal **151** toward the first edge **110**E of the printed circuit board (PCB) **110**. The first tie-bars **142**E and the second tie-bars **152**E may be arranged in the first direction (X direction) along the first edge of the printed circuit board (PCB) **110**.

[0092] The semiconductor module **10** may include the insulation members **160**. Each of the insulation members **160** may include the first region R**1** and the second region R**2** defined by dividing the plane of each insulation member **160**. The first region R**1** of the insulation member **160** may contact the printed circuit board (PCB) **110**. The second region R**2** of the insulation member **160** may contact the signal terminal **141**. The second region R**2** of the insulation member **160** may cover the first end portion of the signal terminal **141** and the first tie-bar **142**E of which a portion is removed by the etch back.

[0093] FIG. **13** is a cross-sectional view of the semiconductor module **10** of FIG. **12** taken along line H-H′ and line I-I′. Section (A) shows a cross-section of the semiconductor module **10** of FIG. **12** taken along line H-H′, and section (B) shows a cross-section of the semiconductor module **10** of FIG. **12** taken along line I-I′.

[0094] Referring to FIG. **13**, in section (A), the signal terminal **141**, the signal wire **143** and the

insulation member **160** may be disposed on the printed circuit board (PCB) **110**. The insulation member **160** may include the first region R**1** covering the printed circuit board (PCB) **110** and the second region R**2** covering the first end portion of the signal terminal **141**. The first region R**1** of the insulation member **160** may be located at a first level, and the second region R**2** of the insulation member **160** may be located at a second level that is higher than the first level. In other words, a surface of the first region is located at a first level relative to the contact surface of the PCB **110**, a surface of the second region is located at a second level relative to the contact surface of the PCB **110**, and the second level is further outward (i.e., along the Z direction) from the contact surface of the PCB **110**. With reference to the contact surface of the printed circuit board (PCB) **110** and the insulation member **160**, the surface of the insulation member **160** may have the first height H**1** in the first region R**1**, and the surface of the insulation member **160** may have the second height H**2** higher than the first height H**1** in the second region R**2**.
[0095] In section (B), the signal terminal **141**, the first tie-bar **142**E of which a portion is removed by the etch back process, and the insulation member **160** may be disposed on the printed circuit board (PCB) **110**. The insulation member **160** may include the first region R**1** covering the printed circuit board (PCB) **110** and the second region R**2** covering cover the first end portion the signal terminal **141** and the first tie-bar **142**E of which a portion is removed by the etch back process.
[0096] FIG. **14** is a perspective view showing that the tab terminals **130** of the region G of the semiconductor module **10** of FIG. **11** are in contact with the socket pin **230** of the socket **20** when the semiconductor module **10** of an embodiment is mounted in the socket **20**.
[0097] Referring to FIG. **14**, the first end portion of the signal terminal **141** and the first tie-bar **142**E of which a portion is removed by the etch back process may be covered by the insulation member **160**, and the ground terminal **151** and the second tie-bar **152**E of which a portion is removed by the etch back process may not be covered by the insulation member **160**. Therefore, when mounting the semiconductor module **10** in the socket **20** in the charged state, the ground terminal **151** or the second tie-bar **152**E that is not cover by the insulation member **160** may contact the socket pin **230** of the socket **20** earlier than the signal terminal **141** of which the first end portion is cover by the insulation member **160**. The charge e accumulated in the socket **20** may move to the ground terminal **151** of the semiconductor module **10** through the socket pin **230** in contact with the ground terminal **151** or the second tie-bar **152**E, move to the via hole **154** via the ground wire **153** through the ground terminal **151**, and exit to the ground through the ground wire layer of the printed circuit board (PCB) **110**. The discharge path may be formed as a path passing through the socket **20**, the socket pin **230**, the ground terminal **151** (or the ground terminal **151** through the second tie-bar **152**E), the ground wire **153**, the via hole **154**, the ground wire layer, and the ground. Therefore, according to the present disclosure, since the discharge path leading to the ground is formed, the charge e accumulated in the socket **20** may be prevented from moving to the semiconductor chip **120**, thereby damaging a circuit within the semiconductor chip **120**, and deteriorating the signal integrity (SI).
[0098] In addition to the above-described content, contents shown in and described with reference to FIG. **5** to FIG. **8** may be equally applied to features with respect to FIG. **11** to FIG. **14**.
[0099] FIG. **15** is a perspective view showing an exemplary variation of the semiconductor module **10** of FIG. **11**. FIG. **16** is an enlarged perspective view showing a region J of the semiconductor module **10** of FIG. **15**.
[0100] Referring to FIG. **15** and FIG. **16**, the semiconductor module **10** may include the insulation member **161**. The insulation member **161** may be disposed on the printed circuit board (PCB) **110**. The insulation member **161** may not cover a ground terminal, a power terminal, plating terminals or no-connection terminals. The insulation member **161** may be spaced apart from the ground terminal, power terminal, plating terminals or no-connection terminals. The insulation member **161** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**.
[0101] The insulation member **161** may include the extension portion **161**E and the protruding

portions **161**P. The extension portion **161**E may extend in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. The extension portion **161**E may be located between the tab terminals **130** and the first edge **110**E of the printed circuit board (PCB) **110**, as illustrated. The extension portion **161**E may contact the printed circuit board (PCB) **110**. The extension portion **161**E may be spaced apart from the tab terminals **130**. The extension portion **161**E may be spaced apart from the signal terminals **141** and the ground terminals **151**.

[0102] The protruding portions **161**P may extend in the second direction (Y direction) from the extension portion **161**E. The protruding portions **161**P may be integrally formed with the extension portion **161**E. The protruding portions **161**P may be disposed with a step in the second direction (Y direction) with respect to the extension portion **161**E (i.e., the protruding portions **161**P extend above the extension portion **161**E along the Y direction). The protruding portions **161**P may be arranged in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. Each of the protruding portions **161**P may contact the printed circuit board (PCB) **110**, the signal terminal **141**, and the first tie-bar **142**E of which a portion is removed by the etch back process. Each of the protruding portions **161**P may cover the first end portion of the signal terminal **141** and the first tie-bar **142**E of which a portion is removed by the etch back process. The first end portion may be defined as a surface that contacts the socket pin **230** when the insulation member **160** is not present, located adjacent to the first edge **110**E of the printed circuit board (PCB) **110**, and having a predetermined area. The protruding portions **161**P may be spaced apart from the ground terminals **151**. Each of the protruding portions **161**P may be spaced apart from a neighboring protruding portion **161**P at a predetermined interval in the first direction (X direction). Each of the protruding portions **161**P may include the first region R**1** and the second region R**2** defined by dividing the plane of the protruding portion **161**P. The first region R**1** may contact the printed circuit board (PCB) **110**. The second region R**2** may contact the first end portion of the signal terminal **141** and the first tie-bar **142**E of which a portion is removed by the etch back process.

[0103] FIG. **17** is a perspective view showing the semiconductor module **10** of still another embodiment. FIG. **18** is an enlarged perspective view showing a region K of the semiconductor module **10** of FIG. **17**.

[0104] Referring to FIG. **17** and FIG. **18**, the semiconductor module **10** may include the insulation members **160**. Each of the insulation members **160** may cover the first end portion of each of the signal terminals **141**.

[0105] The semiconductor module **10** may not include the tie-bars **132** and the tie-bars **132**E of which a portion is removed by the etch back process. In an embodiment, the tab terminals **130** may be formed by performing an electroless plating process without the tie-bars **132** and the tie-bars **132**E of which a portion is removed by the etch back process. The semiconductor module **10** may include the insulation members **160**. Each of the insulation members **160** may include the first region R**1** and the second region R**2** defined by dividing the plane of each insulation member **160**. The first region R**1** of the insulation member **160** may contact the printed circuit board (PCB) **110**. The second region R**2** of the insulation member **160** may contact the signal terminal **141**. The second region R**2** of the insulation member **160** may cover the first end portion of the signal terminal **141**.

[0106] FIG. **19** is a cross-sectional view of the semiconductor module **10** of FIG. **18** taken along line L-L′ and line M-M′. Section (A) shows a cross-section of the semiconductor module **10** of FIG. **18** taken along line L-L′, and section (B) shows a cross-section of the semiconductor module **10** of FIG. **18** taken along line M-M′.

[0107] Referring to FIG. **19**, in section (A), the signal terminal **141**, the signal wire **143** and the insulation member **160** may be disposed on the printed circuit board (PCB) **110**, and in section (B), the signal terminal **141** and the insulation member **160** may be disposed on the printed circuit board (PCB) **110**. The insulation member **160** may include the first region R**1** covering the printed circuit board (PCB) **110** and the second region R**2** covering the first end portion of signal terminal **141**. The first region R**1** of the insulation member **160** may be located at a first level, and the second region R**2** of the insulation member **160** may be located at a second level that is higher than the first level. In other words, a surface of the first region is located at a first level relative to the contact surface of the PCB **110**, a surface of the second region is located at a second level relative to the contact surface of the PCB **110**, and the second level is further outward (i.e., along the Z direction) from the contact surface of the PCB **110** than the first level. With reference to the contact surface of the printed circuit board (PCB) **110** and the insulation member **160**, the surface of the insulation member **160** may have the first height H**1** in the first region R**1**, and the surface of the insulation member **160** may have the second height H**2** higher than the first height H**1** in the second region R**2**.

[0108] FIG. **20** is a perspective view showing that the tab terminals **130** of the region K of the semiconductor module **10** of FIG. **17** are in contact with the socket pin **230** of the socket **20** when the semiconductor module **10** of an embodiment is mounted in the socket **20**.

[0109] Referring to FIG. **20**, the first end portion of the signal terminal **141** may be covered by the insulation member **160**, and the ground terminal **151** may not be covered by the insulation member **160**. Therefore, when mounting the semiconductor module **10** in the socket **20** in the charged state, the ground terminal **151** that is not cover by the insulation member **160** may contact the socket pin **230** of the socket **20** earlier than the signal terminal **141** of which the first end portion is cover by the insulation member **160**. The charge e accumulated in the socket **20** may move to the ground terminal **151** of the semiconductor module **10** through the socket pin **230** in contact with the ground terminal **151**, move to the via hole **154** via the ground wire **153** through the ground terminal **151**, and exit to the ground through the ground wire layer of the printed circuit board (PCB) **110**. The discharge path may be formed as a path passing through the socket **20**, the socket pin **230**, the ground terminal **151**, the ground wire **153**, the via hole **154**, the ground wire layer, and the ground. Therefore, according to the present disclosure, since the discharge path leading to the ground is formed, the charge e accumulated in the socket **20** may be prevented from moving to the semiconductor chip **120**, thereby damaging a circuit within the semiconductor chip **120**, and deteriorating the signal integrity (SI).

[0110] In addition to the above-described content, contents shown in and described with reference to FIG. **5** to FIG. **8** may be equally applied to features with respect to FIG. **17** to FIG. **20**.

[0111] FIG. **21** is a perspective view showing an exemplary variation of the semiconductor module **10** of FIG. **17**. FIG. **22** is an enlarged perspective view showing a region N of the semiconductor module **10** of FIG. **21**.

[0112] Referring to FIG. **21** and FIG. **22**, the semiconductor module **10** may include the insulation member **161**. The insulation member **161** may be disposed on the printed circuit board (PCB) **110**. The insulation member **161** may not cover a ground terminal, a power terminal, plating terminals or no-connection terminals. The insulation member **161** may be spaced apart from the ground terminal, power terminal, plating terminals or no-connection terminals. The insulation member **161** may be disposed along the first edge **110**E of the printed circuit board (PCB) **110**.

[0113] The insulation member **161** may include the extension portion **161**E and the protruding portions **161**P. The extension portion **161**E may extend in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. The extension portion **161**E may be located between the tab terminals **130** and the first edge **110**E of the printed circuit board (PCB) **110**. The extension portion **161**E may contact the printed circuit board (PCB) **110**. The extension portion **161**E may be spaced apart from the tab terminals **130**, as illustrated. The extension portion **161**E may be spaced apart from the signal terminals **141** and the ground terminals **151**.

[0114] The protruding portions **161**P may extend in the second direction (Y direction) from the extension portion **161**E. The protruding portions **161**P may be integrally formed with the extension portion **161**E. The protruding portions **161**P may be disposed with a step in the second direction (Y

direction) with respect to the extension portion **161**E (i.e., the protruding portions **161**P extend above the extension portion **161**E along the Y direction). The protruding portions **161**P may be arranged in the first direction (X direction) along the first edge **110**E of the printed circuit board (PCB) **110**. Each of the protruding portions **161**P may contact the printed circuit board (PCB) **110**, and the signal terminal **141**. Each of the protruding portions **161**P may cover the first end portion of the signal terminal **141**. The first end portion may be defined as a surface that contacts the socket pin **230** when the insulation member **160** is not present, located adjacent to the first edge **110**E of the printed circuit board (PCB) **110**, and having a predetermined area. The protruding portions **161**P may be spaced apart from the ground terminals **151**. Each of the protruding portions **161**P may be spaced apart from the neighboring protruding portion **161**P at a predetermined interval in the first direction (X direction). Each of the protruding portions **161**P may include the first region R**1** and the second region R**2** defined by dividing the plane of the protruding portion **161**P. The first region R**1** may contact the printed circuit board (PCB) **110**. The second region R**2** may contact the first end portion of the signal terminal **141**.
[0115] While this invention has been described in connection with what is presently considered to be practical embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims.

Claims

1. A semiconductor module, comprising: a printed circuit board; a plurality of tab terminals on the printed circuit board, wherein the plurality of tab terminals comprise a plurality of signal terminals and a plurality of ground terminals, wherein the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, wherein each of the plurality of tab terminals extends in a second direction that is transverse to the first direction, and wherein each of the plurality of signal terminals comprises a first end portion adjacent to the first edge of the printed circuit board; and a plurality of insulation members on the printed circuit board, wherein each of the plurality of insulation members covers the first end portion of a respective one of the plurality of signal terminals.
2. The semiconductor module of claim 1, wherein the plurality of ground terminals are spaced apart from the plurality of insulation members.
3. The semiconductor module of claim 1, wherein the plurality of tab terminals further comprise one or more power terminals.
4. The semiconductor module of claim 3, wherein the one or more power terminals are spaced apart from the plurality of insulation members.
5. The semiconductor module of claim 1, wherein the plurality of insulation members comprise a photo solder resist (PSR).
6. The semiconductor module of claim 1, wherein the plurality of insulation members comprise an overcoating material.
7. The semiconductor module of claim 6, wherein: the overcoating material comprises polyurethane acrylate, urethane acrylate, acryl(Acrylic), modified acrylate, or epoxy.
8. The semiconductor module of claim 1, wherein: each of the plurality of insulation members comprises a first region and a second region; the first region contacts the printed circuit board; and the second region contacts the first end portion of a respective one of the plurality of signal terminals.
9. The semiconductor module of claim 8, wherein: a surface of the first region is located at a first level relative to a surface of the printed circuit board; a surface of the second region is located at a second level relative to the surface of the printed circuit board; and the second level is further outward from the surface of the printed circuit board than the first level.

10. The semiconductor module of claim 1, wherein: each of the plurality of signal terminals comprises a first exposed surface, and each of the plurality of ground terminals comprises a second exposed surface; and a distance in the second direction from the first edge of the printed circuit board to the first exposed surface is greater than a distance in the second direction from the first edge of the printed circuit board to the second exposed surface.

11. A semiconductor module, comprising: a printed circuit board; a plurality of tab terminals on the printed circuit board, wherein the plurality of tab terminals comprise a plurality of signal terminals and a plurality of ground terminals, wherein the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, wherein each of the plurality of tab terminals extends in a second direction that is transverse to the first direction, wherein each of the plurality of signal terminals comprises a first end portion adjacent to the first edge of the printed circuit board; and an insulation member on the printed circuit board, wherein the insulation member comprises an extension portion extending in the first direction along the first edge of the printed circuit board, and a plurality of protruding portions extending in the second direction from the extension portion, and wherein each of the plurality of protruding portions covers the first end portion of a respective one of the plurality of signal terminals.

12. The semiconductor module of claim 11, wherein the plurality of ground terminals are spaced apart from the plurality of protruding portions.

13. The semiconductor module of claim 11, wherein the extension portion extends along the first edge of the printed circuit board between the plurality of tab terminals.

14. The semiconductor module of claim 11, wherein the extension portion is spaced apart from the plurality of tab terminals.

15. The semiconductor module of claim 11, wherein: each one of the plurality of protruding portions comprises a first region and a second region; the first region contacts the printed circuit board; the second region contacts the first end portion of a respective one of the plurality of signal terminals.

16. A semiconductor module, comprising: a printed circuit board; a plurality of semiconductor chips on the printed circuit board; a plurality of tab terminals on the printed circuit board, wherein the plurality of tab terminals comprise a plurality of signal terminals electrically connected to the plurality of semiconductor chips, and a plurality of ground terminals connected to a ground, wherein the plurality of tab terminals are arranged in a first direction along a first edge of the printed circuit board, wherein each of the plurality of tab terminals extends in a second direction that is transverse to the first direction and wherein each of the plurality of signal terminals comprises a first end portion adjacent to the first edge of the printed circuit board; a plurality of first tie-bars on the printed circuit board, wherein each of the plurality of first tie-bars extends from a respective one of the plurality of signal terminals; a plurality of second tie-bars on the printed circuit board, wherein each of the plurality of second tie-bars extends from a respective one of the plurality of ground terminals; and a plurality of insulation members on the printed circuit board, wherein each of the plurality of insulation members covers the first end portion and the first tie-bar of a respective one of the plurality of signal terminals.

17. The semiconductor module of claim 16, wherein the plurality of first tie-bars and the plurality of second tie-bars extend in the second direction.

18. The semiconductor module of claim 17, wherein the plurality of first tie-bars and the plurality of second tie-bars extend to the first edge of the printed circuit board.

19. The semiconductor module of claim 16, wherein the plurality of ground terminals are spaced apart from the plurality of insulation members.

20. The semiconductor module of claim 19, wherein the plurality of second tie-bars are spaced apart from the plurality of insulation members.